

Aakkash Muthukumar

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EDUCATION

University of California, San Diego

Master of Science in Computer Science

La Jolla, CA

Sept. 2026 – June 2028

California State University, East Bay

B.S. Computer Science — *Summa Cum Laude* • *Dean's List* (all terms) • GPA: 3.915 / 4.0 Aug. 2023 – May 2026

Hayward, CA

- **Coursework:** Operating Systems, Computer Architecture, Linux Systems Administration, Computer Networks, Data Structures & Algorithms, Analysis of Algorithms, Software Engineering, Artificial Intelligence, Statistics & Probability, Linear Algebra

EXPERIENCE

Automation Intern

Bridge UX Design Studios

May 2025 – July 2025

Bangalore, India

- Designed and shipped an **AI-powered lead-finding system** using n8n workflow orchestration and **LLM-based classification**, parsing structured records from inbound messages and reducing manual processing time by **>70%**
- Authored and delivered an internal **AI-in-UX/UI training workshop** to 5 design stakeholders, translating LLM capabilities into reusable workflows for research, ideation, and client-facing operations — turning non-technical colleagues into hands-on AI users

Course Learning Assistant — STEM Lab

California State University, East Bay

Jan. 2026 – Present

Hayward, CA

- Support 25 CS students with **Python** and **C** programming work — debugging runtime errors, environment configuration, and toolchain setup (**GCC**, **Makefile**, **IDEs**) across assignments and labs
- Coach beginner programmers through systematic debugging strategies: reading compiler output, using **gdb**, and isolating segmentation faults

PROJECTS

Radiography Image Classification | TensorFlow, Keras, MobileNetV2, Streamlit

2024

- Trained a **convolutional neural network** (MobileNetV2 transfer learning) on **~17,000 chest X-ray images** for multi-class radiographic classification, applying **Focal Loss** to address class imbalance in the dataset
- Built a **Streamlit** interpretability dashboard using **Grad-CAM** to visualize model attention regions on input images, supporting explainability for non-technical reviewers

Open-Weight LLM Benchmarking | Python, Ollama, Gemma, GPT-OSS, DeepSeek, Qwen

2025 – 2026

- Designed a reproducible **performance evaluation methodology** comparing **Gemma**, **GPT-OSS**, **DeepSeek**, and **Qwen** on reasoning, code generation, and **inference latency** for offline, on-device deployment
- Documented capability and speed trade-offs into a **model-selection decision framework** for downstream production projects

Custom 64-bit Operating System | C, x86 Assembly, QEMU

Fall 2025

- Built a **64-bit operating system** from scratch in **C** with a hand-written **x86 assembly** bootloader, transitioning through **real mode** → **32-bit protected mode** → **64-bit long mode** during boot
- Configured **GDT** and **page tables** for long-mode operation; boots to an interactive **kernel shell**; developed and iteratively debugged the system under **QEMU**

TECHNICAL SKILLS

Languages: Python, C++, C, Java, JavaScript/TypeScript, x86 Assembly, SQL, Bash, Shell scripting

Cloud & DevOps: Docker, Docker Compose, Kubernetes, GitHub Actions (CI/CD), containerization, automation, microservices, Agile development

Systems & OS Internals: Bootloader programming, memory/resource management, virtual memory, GDT & page tables, system calls, process lifecycle (fork/exec/wait), **Linux** (Arch, Ubuntu, Debian), systemd, CPU architecture (caches, buses, memory hierarchy)

Web & Frameworks: React, Vite, TypeScript, Tailwind CSS, FastAPI, REST APIs, PostgreSQL

ML & Deep Learning: TensorFlow, Keras, NumPy, SciPy, Pandas, scikit-learn, MobileNetV2 (transfer learning), Focal Loss, Grad-CAM; performance benchmarking (inference latency, throughput trade-offs)

Debugging & Tools: gdb, strace, QEMU, journalctl, systemctl, netstat, GCC, Makefile, Git, Claude Code, Cursor, GitHub Copilot